

Department of Engineering

Finite Element Analysis of a Redbull F1 front wing

SRN: 20072078

7ENT1131-0105-2021 – CAE & Applications

07/05/2022

Table of Contents

1.	Introduction	3
2.	Objectives	3
3.	Methodology	4
	a. Mesh Independency Study.....	4
	b. Material optimisation	6
	c. Static Analysis	6
	d. Frequency Analysis.....	7
4.	Results and Analysis.....	7
	a. Static Analysis	7
	b. Frequency Analysis.....	10
5.	Conclusion	11

1. Introduction

Analysis of the Redbull f1 front wing is a comprehensive study that includes mesh optimisation, material selection, static and frequency analysis of the assembly using different materials to obtain a wing which is lightweight and efficient which means less displacement when placed under load.



Fig 1: 2021 Redbull F1 car front wing

Front wings on a formula one car helps in producing downforce and distributing it throughout the body. Normally, they are built using carbon fibre composites that only weighs around 10 to 12kg. Besides being light and strong, they should also be built in such a way that replacement due to damage during a race is easy and consumes less time.

2. Objectives

- Conduct a mesh independency study of the wing assembly with a downforce of 2415N/m^2 using Aluminium as the material and tabulating the outcomes and choosing an optimum mesh size
- Because the model was built of steel, it was redesigned to reduce the overall weight to a more realistic level while keeping the component's structural integrity.
- Material optimisation for the model to help in selection of an optimum material choice for the wing
- Static analysis of the model using optimum mesh size and with different materials to notice the difference in displacement, deformation and stress on the wing and also adding fillets to the edges of the bulkhead to get better flow
- Determining natural frequency of the wing with 8 different materials.

- Concluding the study by examining the tabulated data and choosing the best optimum material

3. Methodology

This study is based on the optimisation techniques used on the industrial level. Although, there are many types of optimisation, Material optimisation seems like the definite approach required to go through with the assessment. Following up with the objectives changing materials of the model decreases its overall weight and enhances its physical properties.

Statistical analysis is particularly chosen in order to estimate the relationship between the variables involved.

a. Mesh Independency Study

Independency study is carried out to determine the optimum mesh size. The aim of this particular study is to determine mesh size and sag values that balances the time taken to finish the simulation with accurate results.

Applying a material such as Aluminium for the wing assembly, the simulations are executed with different combinations of mesh size and sag values. The load on the wing is taken as 2415N/m^2 (respective individual load value).

Material	Mesh size (mm)	Absolute Sag (mm)	Nodes	Elements	Displacement (mm)	Von Mises Stress (N/m ²)	Bulkhead displacement Sensor (mm)
Aluminium	10	1	80573	270039	6.35	6.46E+07	0.190177
	8	0.8	113226	393221	7.11	6.71E+07	0.196645
	6	0.6	161254	555908	7.87	9.22E+07	0.216979
	4	0.4	375974	1382048	8.6	8.84E+07	0.260025
	3.5	0.35	500388	1873313	8.75	8.51E+07	0.250788
	3	0.3	710080	2723570	9.01	8.92E+07	0.263047
	2.5	0.2	1130314	4512984	9.23	1.02E+08	0.295934
	2	0.2	1841378	7501865	9.64	1.41E+08	0.294826

Table 1: Tabulation of data with respect to different mesh size and sag combinations

As shown in Table 1, the variables that influence the selection of optimum mesh size and sag value is displacement in mm as well as Von- Mises stress in N/m^2 with respect to the time taken to compute the results.

Examining below Fig 1 and Fig 2, we can see that, the difference between the values are not nearly equal until the mesh size and sag values are (4, 0.4) and (3.5, 0.35) respectively.

Considering (3.5, 0.35) as the mesh size and sag value, computational time and output of nearly precise results can be obtained. The variation in values from using the mesh size as 4mm and 3.5mm for the study is comparatively very low and also the computation time taken is far less with results that are reliable to continue with the static analysis. Smaller mesh sizes such 2mm or 1mm are not considered because of the computation time the system requires to put out results takes much time than when 3.5mm is considered. Hence, a resolution was made to use the mesh size of 3.5mm and sag values of 0.35mm (which is 10% of the mesh size).

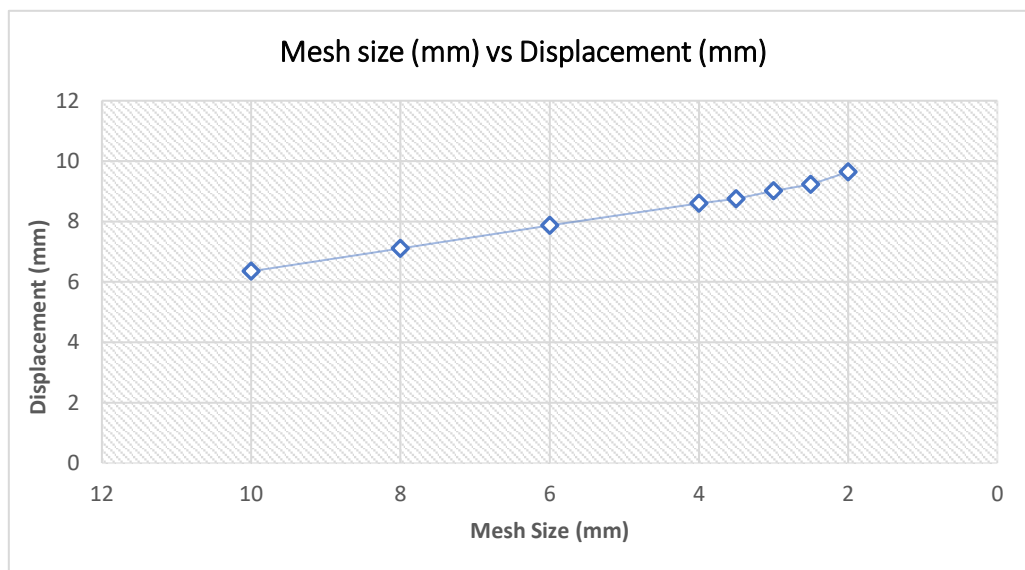


Fig 2: Plot of Mesh size against displacement of the wing in mm

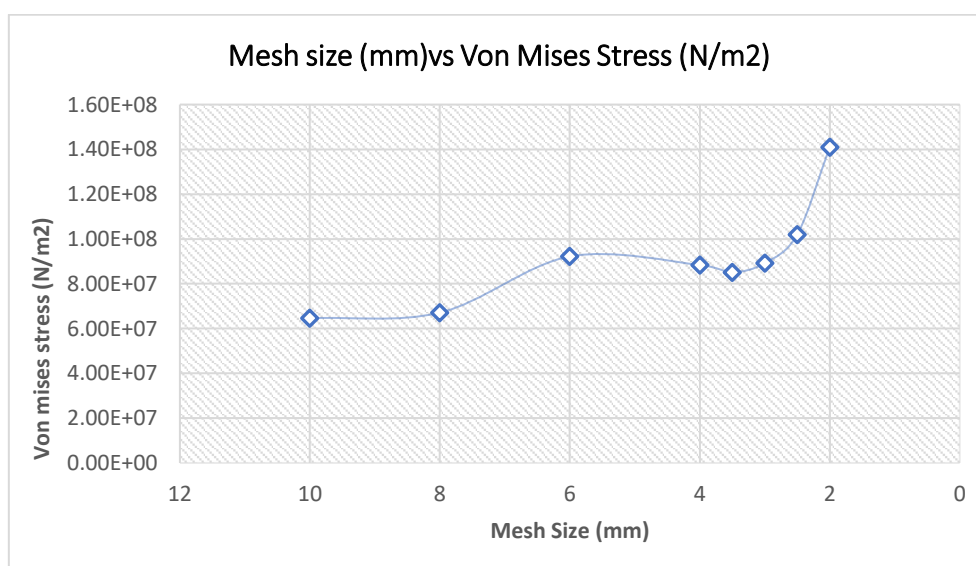


Fig 3: Plot of Mesh size against Von-Mises stress in N/m²

b. Material optimisation

With the mesh size and absolute saga values sorted out, the wing assembly can now be subjected to load using different materials. Materials chosen for the optimisation is mainly specific to either automotive industry or the domain of motorsports.

In below *Table 2*, we can examine the different materials and its properties. Material optimisation is one of the main objectives of this study. Fibreglass and carbon are one of the most used materials in the industry. Although, better materials such as Zylon HM, Spectra 1000 are also taken into account because those materials reach the target results i.e., lightweight with less wing displacement when subjected to a pressure of 2415N/m^2

Materials	Young's Modulus (N/m ²)	Poisson's Ratio	Density (kg/m ³)	Thermal expansion (Kdeg)	Yield Strength (N/m ²)
Fibreglass Grade B	6.90E+10	0.346	2490	2.36E-05	6.57E+10
Fibreglass Grade S	8.70E+10	0.21	2490		4.80E+10
TC250	6.80E+10	0.3	1210		9.50E+07
Carbon (Epoxy composite)	7.00E+10	0.3	1600		9.50E+07
Aluminium 2024T3	7.24E+10	0.33	2810		3.40E+08
Zylon HM	2.70E+11	0.35	1450		5.80E+08
Spectra 1000	1.06E+11	0.3	970		2.35E+09
Kevlar K149	1.79E+11	0.36	1470		1.24E+09

Table 2: Considered materials with its properties

c. Static Analysis

Static analysis is the key step in this study as we have all the initial parameters set and run the simulation in CATIA V5. This type of analysis is mainly done to estimate the relationship between the variables and determine accurate results.

The model is meshed with 3.5mm size and the simulations are run using different materials. *Fig 4* shows the mesh visualization of the wing. The pressure on the wing is set to 2415N/m^2 (individual speed). The surfaces selected for this downforce are the faces which help in creating negative pressure under the car. Bulkhead is clamped so that the wing deformation is equally distributed throughout the body. Other constraints such as coincidence, surface, etc. are applied during the assembly. Furthermore, the material is selected (ref. Table 2) one by one to compute and the data is tabulated for examination

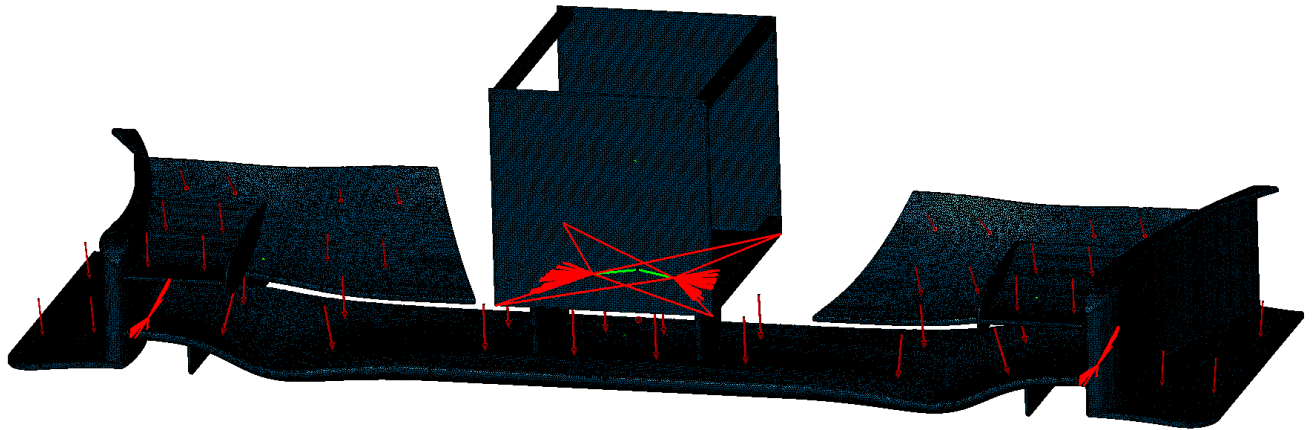


Fig 4: Mesh visualization of the assembly with 3.5mm mesh size and 0.35 sag value

d. Frequency Analysis

Natural frequency is important because when an object vibrates at a frequency equal to its natural frequency, the vibration of the amplitude increases significantly, potentially causing irreparable damage; thus, it is critical to understand the natural frequency. Therefore, considering frequency analysis of the whole assembly with different materials is necessary.

Fig 4 shows the mesh visualization for the static analysis as well as the frequency analysis. The changes done to the process are to the materials and the computation is done as a whole for all the bodies in the assembly.

4. Results and Analysis

a. Static Analysis

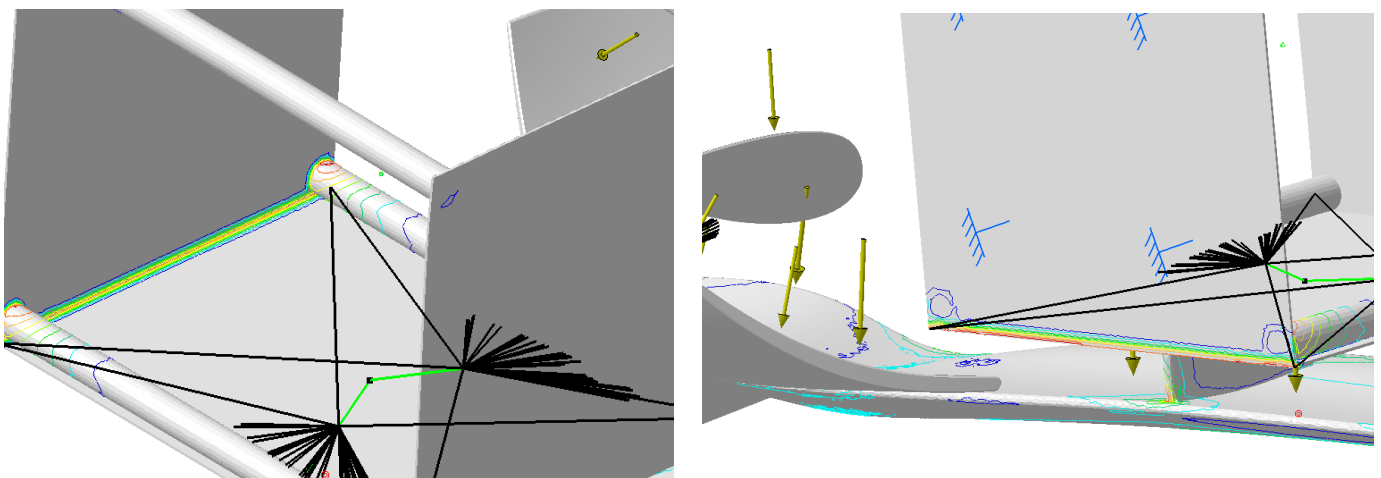


Fig 5: Stress concentration areas

Materials	Pressure (N/m ²)	Von Mises stress (N/m ²)	Displacement (mm)	Mass (kg)	Frequency (Hz) M1	Bulkhead displacement Sensor (mm)
Fibreglass Grade B	2415	8.51E+07	8.88	31.11	10.2289	0.250788
Fibreglass Grade S			6.93	31.11	11.4391	
TC250			9	15.12	14.49	
Carbon (Epoxy composite)			8.74	19.99	12.7848	
Aluminium 2024T3			8.46	35.11	9.83868	
Zylon HM			2.21	18.12	26.5357	
Spectra 1000			5.75	12.12	20.2057	
Kevlar K149			3.37	18.37	21.5048	

Table 3: Static Analysis tabulated data

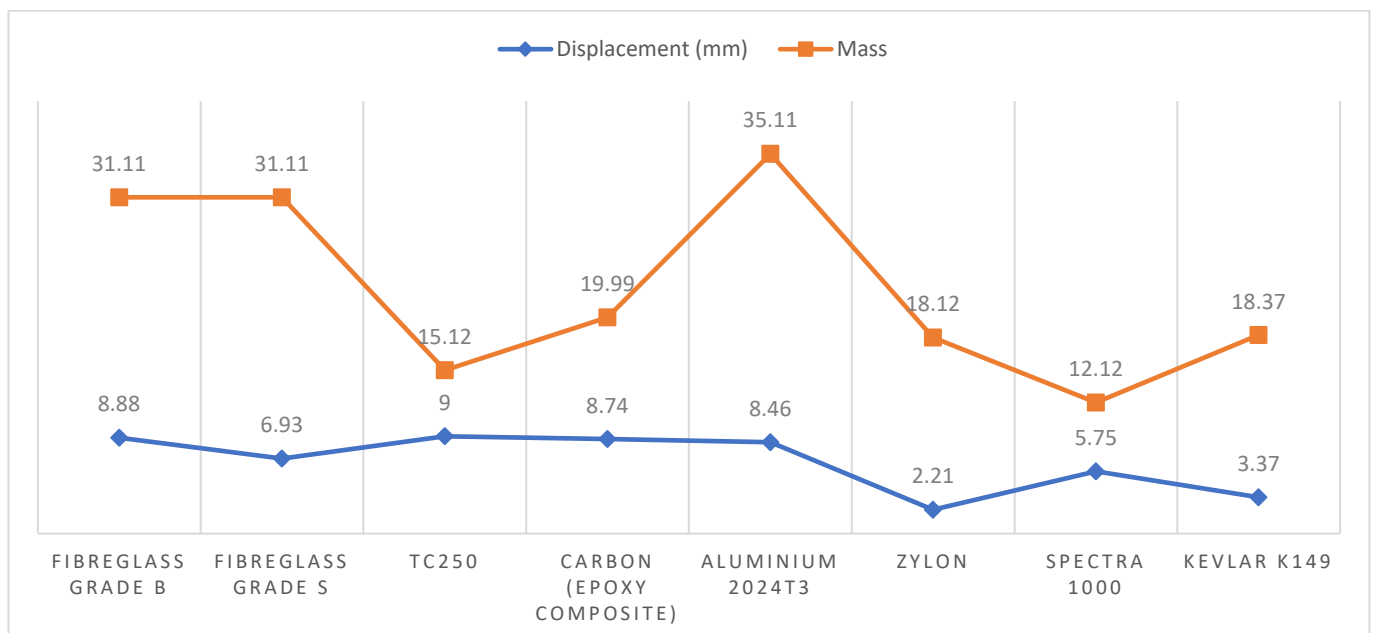


Fig 6: Plot of Mass (kg) and displacement (mm) of respective materials

Above depictions **Table 3 & Fig 5**, summarises the data that the software was able to compute. Investigating the table we can see that the displacement of the wing varies exponentially from 9mm all the way to 3.37mm. Keeping the main objectives of the assignment in mind we have to select the material which has less displacement and which does not weighs more than or equal to 10kg.

ZylonHM would be a great choice for the wing as it has the least displacement. However, the wing weighs a lot more than we want it to. Cost would also be another problem as 50 sq.mt. of Zylon HM costs anywhere between 150GBP to 350GBP. It is expensive.

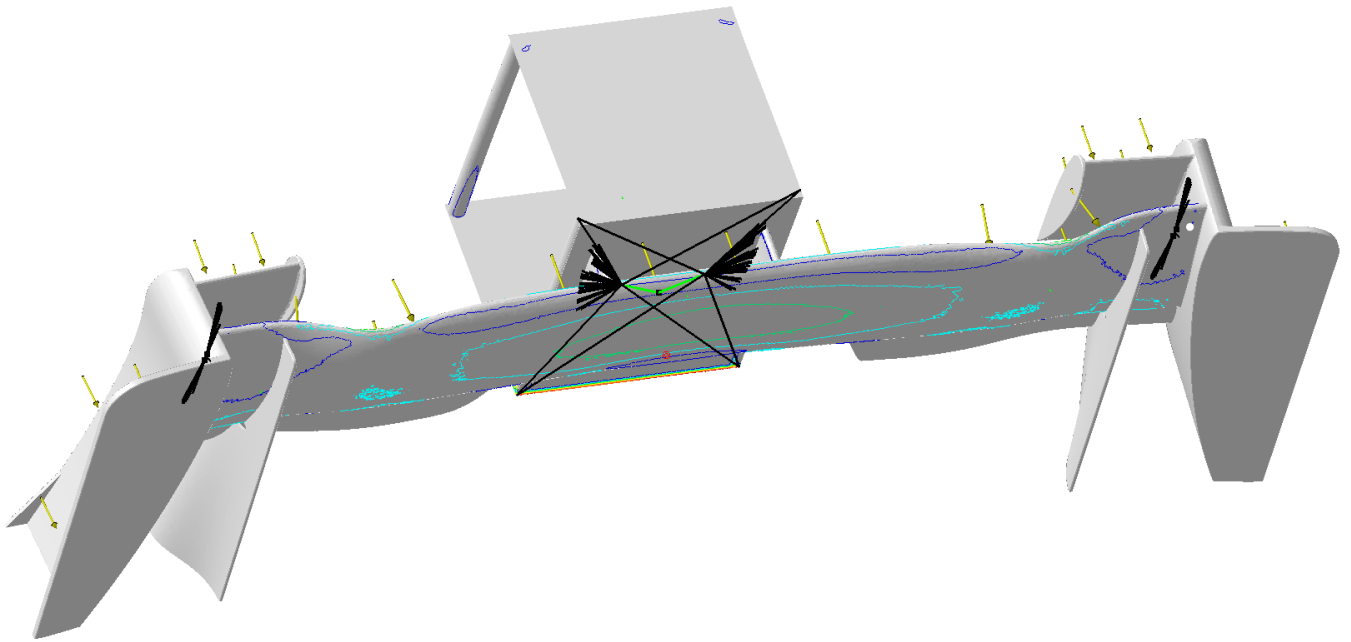


Fig 7: Stress concentration under the wing

KevlarK149 also exhibits the same issue of adding more weight but less displacement.

Spectra1000 is perfect for the job as the displacement is comparatively not higher than 6mm. additionally, the total weight of the wing is 12.12kg. Spectra is inexpensive compared to Zylon and Kevlar

In the beginning of the study the model was made using steel which has high density but low displacement. Comparatively, the aim was to find a material that was stronger than steel but of low density. Spectra1000 is a proper material to be taken into consideration while building an F1 front wing.

Fig 5 shows the stress concentration areas on the wing when subjected to load. Improvements in the form of smoothening sharp edges by adding fillets are done to improve the results. Increase in downward pressure deforms the wing up to a certain degree but adding small changes could improve the results.

b. Frequency Analysis

Frequency Analysis is simulated using the same mesh settings. This has to be done because at natural frequency, the amplitude increases significantly which may cause irreversible damages to the wing.

Table 4, shows the analysis data of natural frequency at 10 different modes in Hz. Furthermore Fig 6 is a plot of the frequency of the respective materials.

Materials	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6	Mode 7	Mode 8	Mode 9	Mode 10
Fibreglass Grade B	10.2289	12.9466	25.1987	28.1539	35.9423	44.169	47.4082	52.656	65.5993	107.578
Fibreglass Grade S	11.4391	14.5801	28.2056	32.0943	40.7335	48.4176	53.8079	59.0665	73.2772	118.581
TC250	14.49	18.4275	35.7455	40.1913	51.2368	62.2774	67.6049	74.7112	93.0746	151.69
Carbon (Epoxy composite)	12.7848	16.2576	31.5314	35.4617	45.2073	54.9461	59.6493	65.9194	82.2027	133.841
Aluminium 2024T3	9.83868	12.4756	24.2364	27.1585	34.6579	42.381	45.7177	50.6807	63.1815	103.315
Zylon HM	26.5357	33.228	62.6679	72.9803	93.1764	101.716	122.899	136.574	156.233	195.766
Spectra 1000	20.2057	25.6492	49.4803	56.0451	71.4475	84.7235	94.2721	104.182	126.734	189.971
Kevlar K149	21.5048	27.0143	51.8142	59.0254	75.3708	87.0073	99.4138	110.626	130.815	194.767

Table 4: Frequency analysis data of all materials for 10 modes in Hz

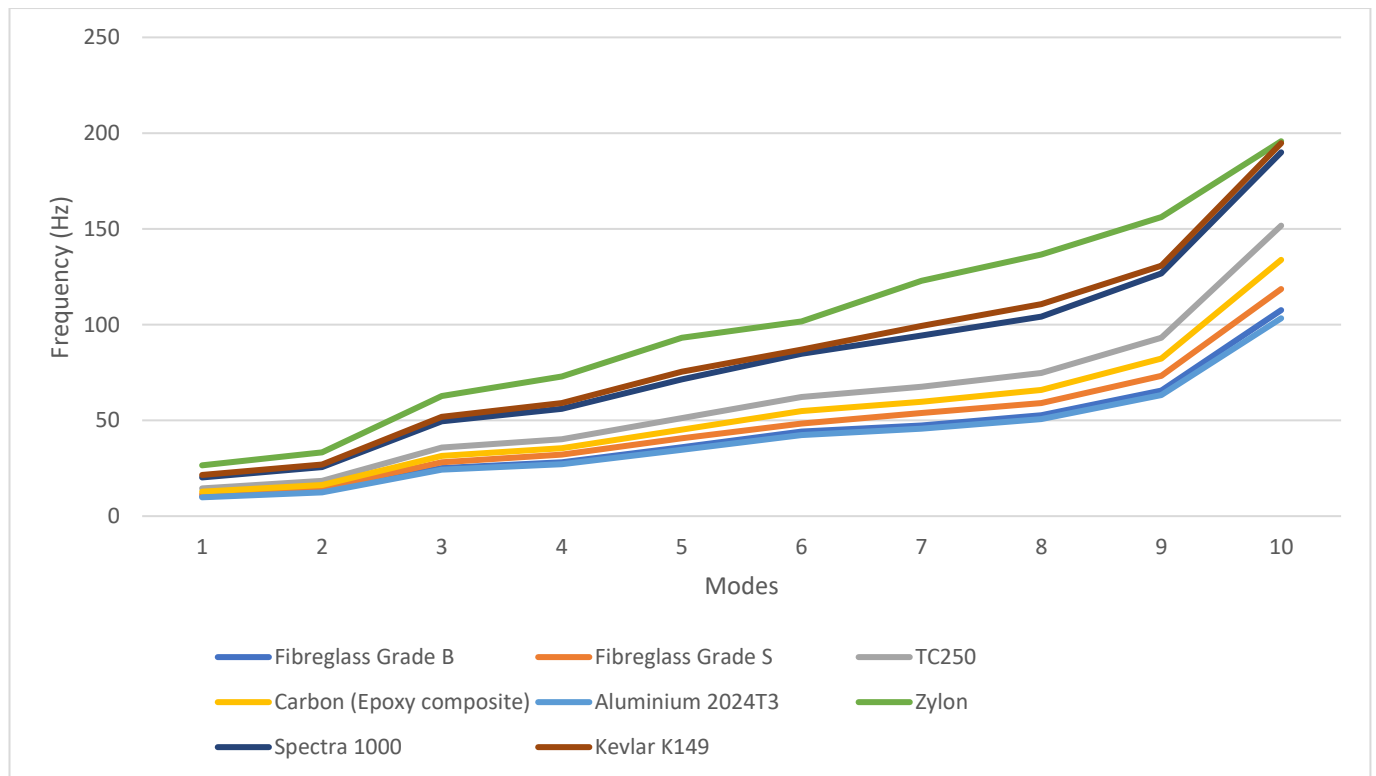


Fig 8: Plot of modes of frequency (Hz) of each material

5. Conclusion

Downforce produced by the front wing is huge and is optimised to not cause drag at high speeds. *Fig 7* clearly shows that most pressure gradient is experienced by the lower section of the wing. Therefore, the material chosen to manufacture these parts must be very strong so that it does not break due to high force, should weigh less to not make the car more heavy and lastly it should be comparatively inexpensive as the current rules and regulations puts cap on the money spent by the teams on their cars.

Upon conducting the simulations we learn that if the wing is made up of Spectra1000 material the displacement is less and also it weighs less than thus improving the cars overall stability. Frequency analysis shows that Spectra1000 is not subjected to damage very easily when compared to other materials